

Utilizing type info in the perf tools

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Data type profiling progress

- Since LPC 2023
  - v6.8 got the initial support
  - v6.10 has full features
  - v6.12 has more improvements
- `perf report`
  - added '**type**' and '**typeoff**' sort keys
- `perf annotate`
  - added '**--data-type**' option



perf mem report -T

- type-level memory access data
  - memory data source, snoop, TLB access
  - hierarchy support may be helpful

```
Samples: 2K of event 'cpu/mem-loads,ldlat=4/P', Event count (approx.): 67777
  Overhead       Samples     Data Type / Memory access
+    19.45%           389     (unknown)
+    12.21%           368     (stack operation)
+     7.68%           439     struct pcpu_hot
-     5.73%            78     struct task_struct
          1.85%              7        RAM hit
          1.29%             48        L1 hit
          1.20%              6        LFB/MAB hit
          1.13%              8        L3 hit
          0.26%              9        L2 hit
```

```
$ perf mem report -T -s type,mem -H
```

perf c2c report

- more focus on cache line level analysis
  - Remote HitM (hit modified) for false sharing
  - Various output fields and sort keys
  - Loads and Stores together
- '**typecln**' sort key (v6.12)
  - per-type cache line offset
  - this can be used by `perf mem report` as well



How to help data layout

- For spatial locality
  - put related fields together
    - may not be intuitive
  - basic block analysis
    - collect memory accesses in the block together
    - a sample could map to (field1, field2, ...)
- What about temporal locality?
  - sample timestamp
    - field level time chart?



Layout suggestion

- Can we automate struct layout?
  - based on the data type profile
  - with pahole
  - with compiler





Issues

- DWARF availability
  - Using line number info?
  - Using BTF?
  - Dumping type info during compilation?
- DWARF quality
  - Location expression
  - LLVM has `-fdebug-info-for-profiling`
  - Verifying instruction tracking
- Other use cases?



Using BTF

- When DWARF is not available
  - no location expression
  - by instruction tracking only
- Supported info
  - function params
  - per-cpu variables
- global variable support is WIP
- no local variable support





What else?

- Any ideas?





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